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COMPUTER

CHAPTER 10

Programming Language

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COMPUTER · CHAPTER 10

Programming Language

ਪ੍ਰੋਗ੍ਰਾਮਿੰਗ ਭਾਸ਼ਾ

IN THIS CHAPTER

- Program Design Tools
- Programming Language
- Types Of Programming Languages
- Low-Level Language
- High-Level Language
- Coding Systems

PROGRAM DESIGN TOOLS

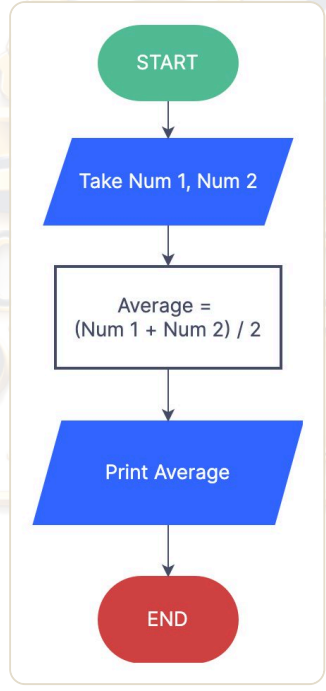
Before writing a program, programmers use certain tools to understand and represent the logic of the program.

1. Flowchart

A **Flowchart** is the pictorial (diagrammatic) representation of an algorithm or program.

Algorithm: A step-by-step set of instructions used to solve a problem.

Simple Flowchart Representation



Exam Point

Flowchart = Graphical Representation of Algorithm

2. DFD (Data Flow Diagram)

A **Data Flow Diagram (DFD)** is a graphical representation that shows how data moves through a system or subsystem.

Example

Student

↓

Registration Form

↓

College Database

↓

Admission Confirmation

Exam Point

DFD shows movement of data, whereas Flowchart shows program logic.

3. Pseudocode (PDL)

Pseudocode is an informal way of writing program logic using simple English statements.

It is also known as PDL (Program Design Language)

Example

BEGIN

Declare num1, num2, sum

Read num1

Read num2 sum = num1 + num2

Print sum

END

Exam Point

Pseudocode = Program logic written in simple English-like statements.

PROGRAMMING LANGUAGE

A **Programming Language** is a medium used to communicate instructions to a computer.

It consists of:

- Keywords
- Symbols
- Rules (Syntax)

through which humans can write instructions that are executed by a computer.

Syntax

Syntax refers to the set of rules that define how statements are written in a programming language.

Example

In C language:

printf("Hello"); must follow proper syntax; otherwise, an error occurs.

TYPES OF PROGRAMMING LANGUAGES

Programming languages are mainly classified into:

```
Programming Languages
│
```

```
├── Low-Level Languages
│       ├── Machine Language
│       └── Assembly Language
│
└── High-Level Languages
        ├── FORTRAN
        ├── ALGOL
        ├── COBOL
        ├── C
        ├── C++
        └── Others
```

1. LOW-LEVEL LANGUAGE

Low-level languages are closer to computer hardware.

Characteristics

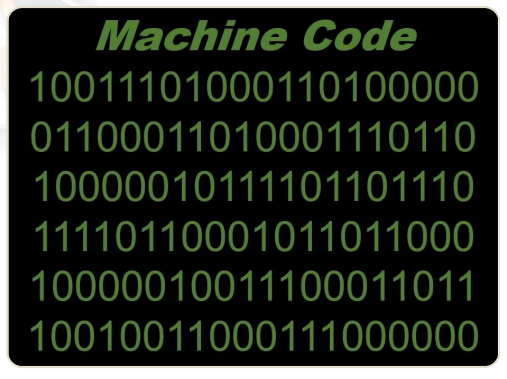
- Machine dependent.
- Faster execution.
- Difficult for humans to understand.
- Requires detailed knowledge of hardware.

A. Machine Language

Machine Language is the lowest-level programming language.

It is also known as:

- Machine Code
- Object Code



Characteristics

- Consists only of binary digits (0 and 1).
- Directly understood by the CPU.
- No translator is required.
- Fastest execution speed.
- Very difficult for humans to write.

Example

10110010

Components of Machine Language Instruction

Component	Meaning
Opcode	Operation to be performed
Operand	Data on which operation is performed

Disadvantages

- Difficult to learn.
- Difficult to debug.
- High possibility of errors.
- Machine dependent.

Exam Point

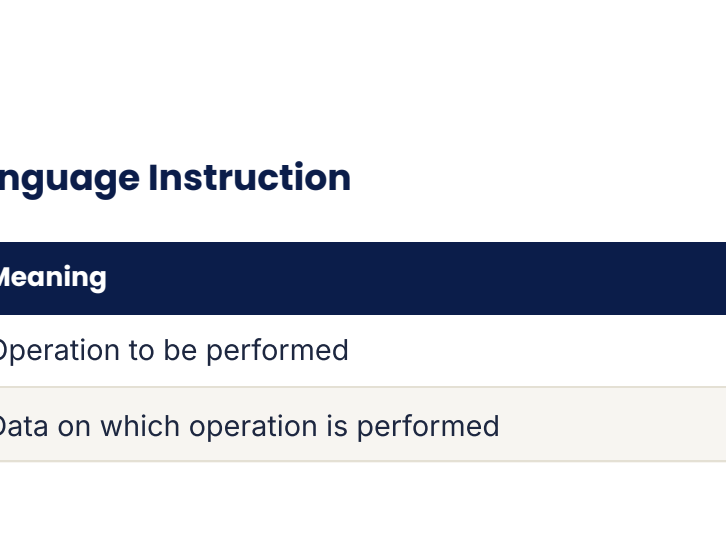
Machine Language is the only language directly understood by the CPU.

B. Assembly Language

Assembly Language was developed to overcome the difficulties of Machine Language.

Instead of binary numbers, it uses mnemonic codes and symbolic addresses.

Examples of Mnemonics



Mnemonic	Meaning
ADD	Addition
SUB	Subtraction
MUL	Multiplication
MOV	Move Data

Translator Used: Programs written in Assembly Language are translated into Machine Language using an Assembler

Advantages

- Easier than Machine Language.
- Easier to remember.

- Easier debugging.

2. HIGH-LEVEL LANGUAGE

A High-Level Language is a machine-independent programming language that is closer to human language.

Programs written in high-level language are called Source Code

Characteristics

- Easy to learn.
- Easy to write.
- Machine independent.
- Requires a translator.

```
print("This program is going to state the person \n")

# set a variable for what each person is like
john = input ("What is John like? ")
sue = input ("What is Sue like? ")
derek = input ("What is Derek like? ")
paul = input ("What is paul like? ")
print("\n")

# Now print them out

print ("John is ",john,"\n")
print ("Sue is ",sue,"\n")
print ("Derek is ",derek,"\n")
print ("paul is ",paul,"\n")
```

Translators Used

Translator	Purpose
Compiler	Converts entire program at once
Interpreter	Converts line by line
Assembler	Converts Assembly Language

Advantages

- Easy coding.
- Easy debugging.
- Portable.
- Faster program development.

IMPORTANT HIGH-LEVEL LANGUAGES

1. FORTRAN (FORmula TRANslation)

Developed By John Backus

Developed For IBM 704 in Year 1956

Use :Scientific and engineering calculations.

Exam Point

FORTRAN was the first widely used High-Level Programming Language.

2. ALGOL(ALGOrithmic Language)

Use :Scientific and mathematical applications.

Importance :Influenced many modern programming languages.

3. COBOL(COmmon Business Oriented Language)

Use:Business and commercial applications.

Example:Banking systems, payroll systems.

4. C Language

Developed By Dennis Ritchie in Year 1972 Developed At Bell Laboratories

Purpose :Development of UNIX Operating System.

Exam Point

C Language was developed by Dennis Ritchie in 1972 at Bell Labs.

5. C++

Developed By Bjarne Stroustrup

Features

- Object-Oriented Programming Language.
- Combines high-level and low-level features.

Important Fact

C++ is often called a **Middle-Level Language** because it supports both low-level and high- level programming features.

6. SHELL

Shell is a command language used with the UNIX operating system.

Exam Point

UNIX operating systems are widely used on web servers.

CODING SYSTEMS

A coding system is used to represent characters, numbers, and symbols in binary form.

1. BINARY CODED DECIMAL (BCD)

BCD is a coding system in which each decimal digit is represented using 4 bits.

Example

Decimal Digit	BCD Representation
0	0000
1	0001
5	0101
9	1001

Example

Decimal Number 59

5 = 0101

9 = 1001

Therefore, 59 in BCD = 0101 1001

Exam Point

BCD uses 4 bits for each decimal digit.

2. ASCII CODE

Full Form :American Standard Code for Information Interchange

ASCII is a character encoding system used to represent letters, digits, and symbols.

Characteristics

- Uses 7 bits.
- Represents 128 characters.
- Widely used in computers.

Examples

Character	ASCII Value
A	65
B	66
a	97
0	48

Important Correction

Original ASCII uses **7 bits** and defines **128 characters**.

Exam Point

ASCII = 7-bit code = 128 characters

3. EBCDIC

Full Form :Extended Binary Coded Decimal Interchange Code

EBCDIC is a character encoding system developed by IBM.

Characteristics

- Uses 8 bits.
- Represents 256 characters.
- Mainly used on IBM mainframe computers.

Exam Point

EBCDIC = 8-bit code

MACHINE LANGUAGE VS ASSEMBLY LANGUAGE VS HIGH- LEVEL LANGUAGE

Feature	Machine Language	Assembly Language	High-Level Language
Representation	0 and 1	Mnemonics	English-like words
Ease of Use	Very Difficult	Moderate	Easy
Translator Required	No	Assembler	Compiler/Interpreter
Speed	Fastest	Fast	Relatively Slow
Machine Dependency	Yes	Yes	No

ASCII VS EBCDIC

Feature	ASCII	EBCDIC
Full Form	American Standard Code for Information Interchange	Extended Binary Coded Decimal Interchange Code
Bits Used	7 Bits	8 Bits
Characters	128	256
Developed By	ANSI	IBM

QUICK REVISION NOTES

- Flowchart = Pictorial representation of algorithm.
- DFD = Shows flow of data through a system.
- Pseudocode = Logic written in English-like statements.
- Machine Language = Directly understood by CPU.
- Assembly Language = Uses mnemonics.
- Translator of Assembly Language = Assembler.
- High-Level Language = Machine independent.
- FORTRAN = First High-Level Language.
- C Language = Dennis Ritchie (1972).
- C++ = Object-Oriented Language.
- BCD = 4 bits per decimal digit.
- ASCII = 7 bits, 128 characters.
- EBCDIC = 8 bits, 256 characters.
- 1 Byte = 8 Bits.

MOST IMPORTANT FACTS FOR EXAMS

1. Machine Language is directly understood by the CPU.
2. Assembly Language uses mnemonics.
3. Assembler converts Assembly Language into Machine Language.
4. FORTRAN was developed by John Backus in 1956.
5. C Language was developed by Dennis Ritchie in 1972.
6. C++ is an Object-Oriented Programming Language.
7. BCD uses 4 bits for each decimal digit.
8. ASCII uses 7 bits and represents 128 characters.
9. EBCDIC uses 8 bits and was developed by IBM.
10. 1 Byte = 8 Bits.